WFLOW-05: PagerDuty & ServiceNow Integration

> **Series:** WFLOW — Workflows and Alert Notifications | **Notebook:** 5 of 10 | **Created:** January 2026 | **Last Updated:** 11/11/2026

Incident Management Automation

Integrate Dynatrace workflows with enterprise incident management platforms. This notebook covers PagerDuty and ServiceNow integration patterns, bi-directional sync, and incident lifecycle management.

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Prerequisites

Requirement	Details
Dynatrace Environment	SaaS with Platform subscription
Permissions	`automation:workflows:write`; `storage:events:read` for the workflow actor (Problem trigger). To create connections: `settings:objects:read`, `settings:objects:write` and `settings:schemas:read` on the connector's schema (for example `app:dynatrace.pagerduty:events-connection`). In **Workflows > Settings > Authorization settings**: `app-settings:objects:read` (both connectors), plus `state:app-states:read`, `state:app-states:write` and `state:app-states:delete` for PagerDuty
PagerDuty	Admin access to create services and integrations
ServiceNow	Admin access or integration user credentials

> **Sources:** Set up PagerDuty Connector (DT docs) — *"ALLOW settings:objects:read, settings:objects:write, settings:schemas:read WHERE settings:schemaId"*; Set up ServiceNow Connector (DT docs) — *"Permissions needed for ServiceNow workflow actions: app-settings:objects:read"*; Problem and event triggers (DT docs).

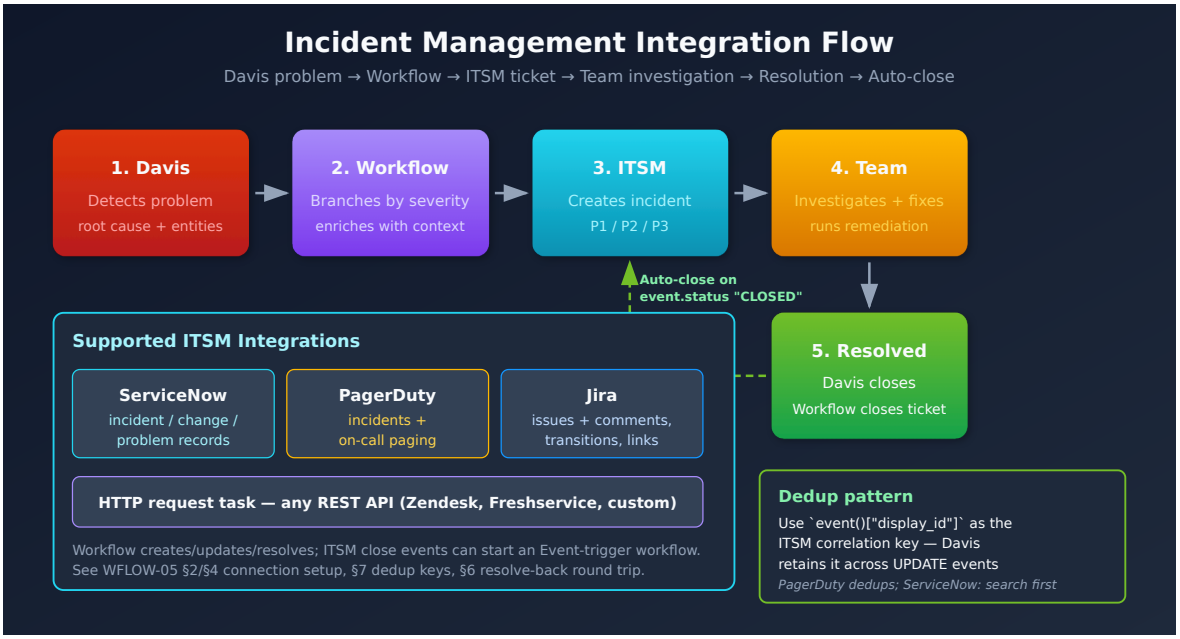
1. Integration Overview

Why Integrate?

Benefit	Description
Single pane of glass	Incidents tracked in one system
On-call management	Leverage PagerDuty/ServiceNow schedules
Audit trail	Complete incident history
Escalation	Built-in escalation policies

Benefit	Description
Metrics	MTTR, incident frequency reporting

Integration Flow



Why a Workflow (vs a Direct Webhook)

Davis can post to a webhook directly, but routing problems through a **workflow** buys four things a raw webhook cannot:

Capability	What the workflow adds
Payload shaping	Map Dynatrace fields to the exact incident schema the target expects, including required and custom fields
Retry + dead-letter	Re-attempt on transient failures; quarantine permanent failures instead of losing them (see §8)
Filtering	Decide which problems become incidents — drop informational events before they reach the ITSM queue
Link-back	Capture the returned incident number and write it onto the Dynatrace problem as a comment (§6)

Reach for the native action first; drop to a raw HTTP action only when you need a field the action doesn't expose.

2. PagerDuty Setup

Creating PagerDuty Integration

1. **In PagerDuty:**
   - Go to **Services** → Select or create service
   - **Integrations** tab → **Add Integration**
   - Select **Events API v2**
   - Copy the **Integration Key** (routing key)

2. **In Dynatrace:**
   - Go to **Settings** → **Connections** → **PagerDuty** and select the **Events API** tab
   - Select **Connection**
   - Paste the Integration Key into **Routing key**
   - Name: `pagerduty-production`

> **Two connection types.** The steps above create an **Events** connection (a routing key), which is what **Send event** uses: *"The Send event action uses the PagerDuty Events API v2 and requires a separate Events connection configured with a routing key."* **Create an incident** and the list actions use a different connection that holds a PagerDuty REST API key. Neither connection can stand in for the other.
>
> **Sources:** Set up PagerDuty Connector (DT docs) — *"Use the PagerDuty Connector with a PagerDuty API key to automate incident creation and retrieve on-call information via the PagerDuty REST API."*

Multiple Services Pattern

Create separate connections for different routing:

Connection Name	PagerDuty Service	Use
`pagerduty-prod-critical`	Production-Critical	P1 incidents
`pagerduty-prod-standard`	Production-Standard	P2-P4 incidents
`pagerduty-platform`	Platform-Infrastructure	Infra issues

3. PagerDuty Workflow Tasks

PagerDuty has no separate resolve or acknowledge action. One action, **Send event** (`dynatrace.pagerduty:send-event`), covers the whole alert lifecycle: its **Event action** input is `trigger` , `acknowledge` or `resolve` . Send every event about one problem with the same deduplication key, through the same Events connection, so PagerDuty updates the existing alert instead of opening a new one.

Trigger an Alert

Input keys below follow an exported workflow (the editor labels them Event action, Severity, Summary, Source, Component, Group, Class, Custom details, Deduplication key); export your own workflow once to confirm them.

```
name: trigger_pagerduty_alert
action: dynatrace.pagerduty:send-event
input:
  connectionId: "{{ connection('app:dynatrace.pagerduty:events-connection',
'pagerduty-production') }}"
  eventAction: trigger
  severity: '{{ {1: "critical", 2: "error", 3: "warning", 4:
"info"}.get(event().get("event.severity") | int(5), "info") }}'
  summary: "[{{ event()['event.category'] }}] {{ event()['event.name'] }}"
  source: "dynatrace"
  # Root-cause name from the Smartscape field, falling back to the classic ID
  component: &gt;-
    {% set rc = event().get("root_cause.smartscape_entity") %}{{ rc["name"] if
rc else (event().get("root_cause_entity_id") or "unknown") }}
  group: ""                    # left empty, as in Dynatrace's template;
problems carry no management_zones
  eventClass: "{{ event()['event.category'] }}"
  customDetails: |             # a JSON object, written as text
    {
      "problem_id": "{{ event()['display_id'] }}",
      "problem_url": "{{ problem_link() }}",
      "affected_entities": "{{ (event().get('smartscape.affected_entities') or
[]) | map(attribute='name') | join(', ') or event()['affected_entity_ids'] |
join(', ') }}",
      "root_cause": "{{ event().get('root_cause_entity_id', 'N/A') }}",
      "start_time": "{{ event()['event.start'] }}"
    }
  dedupKey: "dynatrace-{{ event()['display_id'] }}"
```

> **Deprecated problem fields.** *"root_cause_entity_id and root_cause_entity_name are deprecated in favor of root_cause.smartscape_entity , and affected_entity_ids in favor of smartscape.affected_entities ."* The classic fields still work and are better populated today: over 7 days on the validation tenant (10/06/2026), `affected_entity_ids` was set on 4,244 of 4,244 problems and `smartscape.affected_entities` on 3,332; `root_cause_entity_id` on 850 and `root_cause.smartscape_entity` on 150. The mapping above therefore reads the Smartscape field first and falls back to the classic one. Dynatrace's own PagerDuty template reads only the Smartscape fields and leaves Group empty.
>
> **Sources:** Upgrade guide — alert notification (DT docs), Send problem as event to PagerDuty template (DT docs) — *"Group : The template leaves this field empty."*

Resolve the Alert

Run when the problem closes:

```
name: resolve_pagerduty_alert
action: dynatrace.pagerduty:send-event
input:
  connectionId: "{{ connection('app:dynatrace.pagerduty:events-connection',
'pagerduty-production') }}"
  eventAction: resolve
  severity: info
  summary: "Resolved in Dynatrace: {{ event()['event.name'] }}"
  source: "dynatrace"
  dedupKey: "dynatrace-{{ event()['display_id'] }}"
```

Acknowledge the Alert

```
name: acknowledge_pagerduty_alert
action: dynatrace.pagerduty:send-event
input:
  connectionId: "{{ connection('app:dynatrace.pagerduty:events-connection',
'pagerduty-production') }}"
  eventAction: acknowledge
  severity: info
  summary: "Acknowledged from Dynatrace: {{ event()['event.name'] }}"
  source: "dynatrace"
  dedupKey: "dynatrace-{{ event()['display_id'] }}"
```

Dynatrace's own *Send problem as event to PagerDuty* template does the same in a single task: it sets the event action to `resolve` when the problem's status is `CLOSED`.

Opening an Incident Instead of an Alert

When you need a PagerDuty *incident* opened directly, with a service ID, priority or assignee, use **Create an incident** (`dynatrace.pagerduty:create-incident`) with the REST API connection. Its inputs differ from Send event (From, Title, Service ID, Priority ID, Urgency, Incident key), and it has no resolve counterpart in the connector.

> **Sources:** PagerDuty Connector actions (DT docs) — *"Trigger, acknowledge, or resolve an alert in PagerDuty using the Events API v2"*, *"Creates an incident in your PagerDuty environment for a service."*; Send problem as event to PagerDuty template (DT docs).

4. ServiceNow Setup

Creating ServiceNow Connection

1. **In ServiceNow:**
   - Create an integration user, or an OAuth app
   - Give it the rights the connector's actions use: *"Search, create and update incidents (table incident)"*, read categories and subcategories (table `sys_choice`, elements `category` and `subcategory`), read assignment groups (table `sys_user_group`), and read resolution codes (table `sys_choice`, element `close_code`)
   - Note instance URL: `https://your-instance.service-now.com`
2. **In Dynatrace:**
   - Go to **Settings** → **Connections** → **ServiceNow**
   - Select **Connection**
   - Enter:
     - Instance URL
     - Basic-auth username/password **or** OAuth client ID/secret
     - Name: `servicenow-production`
3. **Allow the outbound calls.** If the actions fail with `Blocked request to '…' (host not in allowlist)`, add the instance host under **Settings** → **General** → **External**

requests. That is the error the ServiceNow actions returned on the validation tenant (10/06/2026) for a host that was not listed. If the instance accepts only allow-listed source IPs, **WFLOW-94** covers sending connector calls from a static egress IP through EdgeConnect.

Authentication Options

The ServiceNow connection supports two authentication methods — these are the only two the native connector accepts:

Method	Credentials	Best For
Basic Authentication	Username + password of a ServiceNow integration user	Quick setup, development
OAuth Client Credentials	Client ID + client secret	Production — avoids storing a user password

> There is no API-key option for this connector. Store the chosen credential in the connection, never inline in a workflow task.
>
> **Sources:** Set up ServiceNow Connector (DT docs) — *"Read resolution codes (table sys_choice , element close_code)"*, *"Either use basic authentication or OAuth client credentials."*

5. ServiceNow Workflow Tasks

The native ServiceNow connector exposes these operations, selected **by name** in the workflow builder. Create Incident is `dynatrace.servicenow:snow-create-incident`, Search incidents is `dynatrace.servicenow:snow-search-incidents`, and Comment on an incident is `dynatrace.servicenow:snow-comment-on-incident`, in Dynatrace's workflow samples. The other operations are shown by name only (`# Operation:`): export a workflow that uses one to read its identifier and input names, rather than guessing them:

Operation	Use
Create Incident	Open an incident from a Davis problem
Resolve incident	Close the incident when the problem closes

Operation	Use
Comment on an incident	Append work notes / status updates
Update record	Change fields on any table record (incidents included)
Search incidents	Look up an existing incident (e.g., by correlation ID)
Get Groups	Resolve assignment-group sys_ids
Create a vulnerability item	Open a VR item (security workflows)
Create Comment	Create a comment on an entry in any table
Create Record	Create a new record in a specified table
Search	Generic search of any table

> **SaaS 1.350 (rollout start 11/04/2026, staged tenant rollout): Send ITOM Event.** *"ServiceNow Connector now includes the Send ITOM Event workflow action. Use it to send Dynatrace problems to ServiceNow IT Operations Management (ITOM) as events. A ready-made workflow template, Send problem as ITOM event to ServiceNow, helps you get started."* The action *"Send an event to ServiceNow Event Management using the ITOM inbound web service. Requires an ITOM events connection."* Verify the action has reached your tenant; until it does, the incident operations above and the ALERT-04 ladder remain the working path.
>
> **Sources:** ServiceNow Connector actions (DT docs), What's new in Dynatrace SaaS 1.350 (DT docs) — *"The template creates a ServiceNow alert when a problem opens and closes that alert when the problem closes."*
>
> **Required fields for Create Incident:** `Category`, `Subcategory`, `Impact`, `Urgency`, and `Assignment Group` are **required** — a create call missing any of them fails. `Correlation ID` is optional but strongly recommended: set it to the Dynatrace problem ID so a **Search incidents** task can find the incident again. Create Incident does not check for an existing incident (it is a plain `POST /api/now/v2/table/incident`), so the deduplication is yours: search first, and create only when nothing is found.

Find or Create the Incident

Search for the problem's correlation ID first, then create only if the search came back empty. This is the pattern Dynatrace's own ServiceNow sample uses.

```
search_incident:
  name: search_incident
  action: dynatrace.servicenow:snow-search-incidents
  predecessors: []
  input:
    connectionId: "{{ connection('app:dynatrace.servicenow:connection',
'servicenow-production') }}"
    sysparmQuery: "correlation_id=DT-{{ event()['display_id'] }}"
    sysparmLimit: "1"
```

```
name: create_snow_incident
action: dynatrace.servicenow:snow-create-incident
predecessors:
  - search_incident
conditions:
  states:
    search_incident: OK
  custom: '{{ event()["event.status"] == "ACTIVE" and
result("search_incident") | length == 0 }}'
  else: SKIP
input:
  connectionId: "{{ connection('app:dynatrace.servicenow:connection',
'servicenow-production') }}"
  shortDescription: "[Dynatrace] {{ event()['event.name'] }}"
  description: |
    A problem has been detected by Davis.

    Problem Details:
    ================
    Problem ID: {{ event()['display_id'] }}
    Category: {{ event()['event.category'] }}
    Status: {{ event()['event.status'] }}
    Start Time: {{ event()['event.start'] }}

    Affected Entities:
    {{ event()['affected_entity_ids'] | join('\n') }}{# deprecated field; see
§3 #}

    Root Cause Entity:
    {{ event().get('root_cause_entity_id', 'Pending analysis') }}

    View in Dynatrace:
    {{ problem_link() }}
  impact: '{{ {1: 1, 2: 1, 3: 2, 4: 3}.get(event().get("event.severity") |
int(5), 3) }}'
  urgency: '{{ {1: 1, 2: 1, 3: 2, 4: 3}.get(event().get("event.severity") |
int(5), 3) }}'
  category: "software"           # a choice value from your instance's
sys_choice list
  subCategory: "application"
  group:                         # the assignment group, picked in the editor
    id: "&lt;sys_id of Platform Engineering&gt;"
    displayName: "Platform Engineering"
  caller: "dynatrace.integration"
  correlationId: "DT-{{ event()['display_id'] }}"
```

> **Sources:** ServiceNow Connector actions (DT docs) — *"Creates an incident in your ServiceNow instance."*; wftpl_sample_servicenow_incident_man.yaml (Dynatrace GitHub) — *"action: dynatrace.servicenow:snow-create-incident"*; wftpl_core_journey_auto-remediation.yaml (Dynatrace GitHub) — *"action: dynatrace.servicenow:snow-comment-on-incident"*.

Severity → Impact / Urgency. Dynatrace's unified `event.severity` is an integer **1–5** (1 Critical, 2 Major, 3 Minor, 4 Warning, 5 Informational; see WFLOW-04 §3 and AIOPS-03), and it is what the Davis problem and `fetch events` carry. ServiceNow `Impact` / `Urgency` run **1 (High) … 3 (Low)**, so the Dynatrace scale compresses:

`event.severity`	ServiceNow Impact / Urgency
1 (Critical)	1 (High)
2 (Major)	1 (High)
3 (Minor)	2 (Medium)
4 (Warning) — events only, never on a problem	3 (Low)
5 (Informational)	skip — usually no incident

> **What `event()` actually carries.** The problem record has `event.severity` on the 1–5 scale, not `"CRITICAL"`-style labels. Grail stores it as a number; in the `event()` payload it may arrive as a string, which is why the mappings above convert it with `| int`. It is `stable` in the semantic dictionary (read 11/11/2026), and from SaaS 1.348 it is no longer defaulted (WFLOW-04 §3), so `int(5)` sends an unset severity to the lowest tier. The same applies to the PagerDuty severity mapping in §3.

Comment on an Incident

```
name: comment_snow_incident
action: dynatrace.servicenow:snow-comment-on-incident
predecessors:
  - search_incident
conditions:
  states:
    search_incident: OK
  custom: '{{ result("search_incident") | length &gt; 0 }}'
  else: SKIP
input:
  connectionId: "{{ connection('app:dynatrace.servicenow:connection',
'servicenow-production') }}"
  # The incident number (INC...), not the correlation ID or the sys_id
  number: '{{ result("search_incident")[0].number }}'
  comment: |
    [Automated Update from Dynatrace]
    Problem updated at {{ now() }}
    Current status: {{ event()['event.status'] }}
```

Resolve Incident

Resolve incident targets an incident by number, not by correlation ID, and all three of its inputs are required: *"Incident Number The number of the incident to resolve."*, *"Resolution Notes Add notes for the resolution of the incident."*, *"Resolution Code The close code for the resolution."* Pick the resolution code from the editor's list: the codes are read from your instance (`sys_choice` , element `close_code`), so a hard-coded `"Resolved"` may not exist there.

```
name: resolve_snow_incident
# Operation: "Resolve incident" (export a workflow to read its action
identifier)
predecessors:
  - search_incident
conditions:
  states:
    search_incident: OK
  custom: '{{ event()["event.status"] == "CLOSED" and
result("search_incident") | length &gt; 0 }}'
  else: SKIP
input:
  connectionId: "{{ connection('app:dynatrace.servicenow:connection',
'servicenow-production') }}"
  number: '{{ result("search_incident")[0].number }}'   # Incident Number
  resolutionNotes: |                                    # Resolution Notes
    Problem automatically resolved in Dynatrace.
    Resolution time: {{ event().get('event.end', now()) }}
  resolutionCode: "&lt;a close code from the editor's list&gt;"   # Resolution
Code
```

> **Sources:** ServiceNow Connector actions (DT docs); wftpl_core_journey_auto-remediation.yaml (Dynatrace GitHub) — reads the incident number from `result("search_incident")[0].number`.

6. Bi-Directional Sync

The native ServiceNow **workflow action is one-way** — it creates, comments on, resolves, and updates ServiceNow records, but ServiceNow-side changes do not flow back automatically. There are two ways to close the loop:

- **DIY polling (shown below)** — a scheduled workflow queries ServiceNow for Dynatrace-correlated incidents and writes their status back onto the Dynatrace problem. Portable and works today, but you own the polling cadence and rate budget.
- **Native bi-directional (ITOM)** — the ServiceNow-side Dynatrace app / ITOM Event Management ingests Dynatrace events and synchronises incident state both ways. This is the richest option for ITOM-centric shops; the **ALERT-04** notebook lays out the full maturity ladder and when to climb to it.

Sync Status from ServiceNow to Dynatrace (DIY polling)

Use a scheduled workflow to check incident status:

```
import { credentialVaultClient } from '@dynatrace-sdk/client-classic-
environment-v2';

const SNOW_URL = 'https://your-instance.service-now.com';   // add to Settings
&gt; External requests

export default async function () {
  // A JavaScript task does not receive a connection object; keep the
credential in the Credential Vault.
  // Required: the workflow actor has the credential-vault:entries:read
permission (for an actor that is
  // not a service user, also turned on under Secondary permissions in the
Workflows authorization settings),
  // the AppEngine scope selected, "Allow access without app context" turned
on, and access for the
  // workflow actor - otherwise this call is denied.
  const cred = await credentialVaultClient.getCredentialsDetails({ id:
'CREDENTIALS_VAULT-XXXXXXXXXXX' });
  // Query ServiceNow for Dynatrace-related incidents. Encoded queries join
field, operator
  // and value with no spaces: correlation_idSTARTSWITHDT-
  const response = await fetch(
    `${SNOW_URL}/api/now/table/incident?` +
    `sysparm_query=${encodeURIComponent('correlation_idSTARTSWITHDT-
^state!=6')}`,
    {
      headers: {
        'Authorization': `Basic ${btoa(cred.username + ':' + cred.password)}`,
        'Accept': 'application/json'
      }
    }
  );

  const incidents = await response.json();

  // Process each incident
  for (const incident of incidents.result) {
    const displayId = incident.correlation_id.replace('DT-', '');   // P-...,
the display ID
    // problemsClient.createComment needs the problem ID (event.id), not the
display ID:
    // look it up in dt.davis.problems by display_id, then post the SNOW
status as a comment
    // ...
  }
```

```
  return { processed: incidents.result.length };
}
```

> **Sources:** Run JavaScript action (DT docs) — *"The workflow actor has the credential-vault:entries:read permission. If the actor isn't a service user, credential-vault:entries:read is also turned on under Secondary permissions in the Workflows authorization settings"* and *"The AppEngine scope is selected for the credential. Allow access without app context is turned on for the credential. The workflow actor has access to the credential."*

Write the Incident Number onto the Problem (link-back)

Create Incident returns the new incident's number. A downstream task can read it directly as `{{ result("create_snow_incident").number }}`, so no extra task is needed just to store it. To give responders a path from the Dynatrace problem to the ticket, post the number as a problem comment:

```
link_back:
  name: link_back
  action: dynatrace.automations:run-javascript
  predecessors:
    - create_snow_incident
  conditions:
    states:
      create_snow_incident: SUCCESS
    else: SKIP
  input:
    script: |
      import { execution, result } from '@dynatrace-sdk/automation-utils';
      import { problemsClient } from '@dynatrace-sdk/client-classic-
environment-v2';

      export default async function () {
        const ev = (await execution()).event();                // the Problem
trigger payload
        if (!ev) throw new Error('No event context - run this from a Problem
trigger');
        const incident = await result('create_snow_incident');
        await problemsClient.createComment({
          problemId: ev['event.id'],
          body: { message: `ServiceNow incident ${incident.number}`, context:
'ServiceNow' },
        });
        return { incident_number: incident.number };
      }
```

Dynatrace's own ServiceNow sample passes `event()["event.id"]` as the `problemId` of the problems API in a Problem-triggered workflow; the comment call takes the same ID.

> **Deprecation — Dynatrace API 1.349 (staged tenant rollout from 10/27/2026).** The API 1.349 changelog (DT docs) marks the Environment API v2 `/problems` endpoints deprecated, including *"POST /problems/{problemId}/comments Deprecated"*, and names no successor. Verify it has reached your tenant; until then the current guidance remains the working path.
>
> **Sources:** wftpl_sample_servicenow_incident_man.yaml (Dynatrace GitHub) — *"{problemId: davisEvent['event.id'], fields: 'impactAnalysis'}"*.

7. Incident Lifecycle Patterns

Complete Lifecycle Workflow

Handle open, update, and close events:

In the editor, set the Problem trigger's **Problem state** to *active or closed*: *"active or closed : Starts when the problem opens and again when it closes."* Without it the workflow never runs on close, and the resolve task never fires. Exported, the trigger and tasks look like this:

```
title: incident-lifecycle-management
trigger:
  eventTrigger:
    isActive: true
    triggerConfiguration:
      type: davis-problem
      value:
        triggerOn: open-and-close     # UI: Problem state = active or closed
        categories:
          availability: true
          error: true
          slowdown: true
          resource: true
          custom: true
        entityTagsMatch: all
        entityTags:
          env:
            - prod

tasks:
  # CREATE: when the problem opens
  trigger_alert:
    name: trigger_alert
    action: dynatrace.pagerduty:send-event
    predecessors: []
    conditions:
      states: {}
      custom: '{{ event()["event.status"] == "ACTIVE" }}'
      else: SKIP
    input:
      connectionId: "{{ connection('app:dynatrace.pagerduty:events-
connection', 'pagerduty-production') }}"
      eventAction: trigger
      source: dynatrace
      severity: critical
      summary: "{{ event()['event.name'] }}"
      dedupKey: "dynatrace-{{ event()['display_id'] }}"

  # RESOLVE: when the problem closes
  resolve_alert:
    name: resolve_alert
    action: dynatrace.pagerduty:send-event
    predecessors: []
    conditions:
      states: {}
      custom: '{{ event()["event.status"] == "CLOSED" }}'
      else: SKIP
```

```
    input:
      connectionId: "{{ connection('app:dynatrace.pagerduty:events-
connection', 'pagerduty-production') }}"
      eventAction: resolve
      source: dynatrace
      severity: info
      summary: "Resolved in Dynatrace: {{ event()['event.name'] }}"
      dedupKey: "dynatrace-{{ event()['display_id'] }}"
```

`triggerOn` is the current key in the `@dynatrace-sdk/client-automation` typings, where `onProblemClose` is marked deprecated; older exports, including Dynatrace's ServiceNow sample, still carry `onProblemClose`.

> **Sources:** Problem and event triggers (DT docs); wftpl_sample_servicenow_incident_man.yaml (Dynatrace GitHub) — the exported `triggerConfiguration` shape.

Deduplication Strategy

Platform	Dedup Field	Pattern
PagerDuty	`dedupKey`	`dynatrace-{problem_id}`
ServiceNow	`correlation_id`	`DT-{problem_id}` — run **Search incidents** on it first and create only if none is found (§5)

PagerDuty deduplicates on the key itself, so a repeated `trigger` updates the open alert. ServiceNow's Create Incident does not: the search-before-create step is what prevents a duplicate.

8. Production Hardening

An incident integration sits on the critical path: when it silently fails, problems open in Dynatrace but never reach the on-call queue. Harden it before you depend on it.

Failure Handling

Decide explicitly what happens on each failure mode rather than letting the workflow fail open:

Failure	Handling
ServiceNow **5xx** (server error)	Turn on the task's **Retry on error** (fixed count and delay, e.g. 3 × 60 s)
ServiceNow **4xx** (bad payload / auth)	Retry cannot tell it apart from a 5xx; a follow-up task with state condition **error** writes the payload to a dead-letter store and alerts the team
Timeout (slow response)	Covered by Retry on error; surface a final failure on the Dynatrace problem as a comment
Rate limit reached	Log dropped events and continue; alert if the sustained drop rate crosses a threshold
DNS / connection failure	Retry; if persistent, alert via a secondary channel (e.g., a Slack action)
Auth token expired	Re-fetch the credential once per run from the connection/secret store; alert ops if it recurs

Retry on error has a fixed count and a fixed delay, with no backoff, and it retries every failure regardless of status code: *"Number of retries , by default, is set to two. Delay between retries (seconds) , by default, is set to 30." "If none of the action retries are successful, the task will end in an error state."* Branch on that error state for the dead-letter path.

> **Sources:** Build workflows — Task options (DT docs).

Dead-Letter Queue

Retries cover transient failures; **permanent** failures (4xx, schema mismatch) must not vanish. Write the failed payload to a durable store — a dedicated Grail bucket or an external queue — with a retention window (e.g., 30 days) so an operator can inspect and **replay** it after fixing the root cause. Without a dead-letter store, a malformed-payload bug means silently lost incidents with no audit trail.

Link Back to the Dynatrace Problem

Capture the ServiceNow incident number from the create response and write it onto the originating problem as a comment (the link-back task in §6). This gives responders a one-

click path from the Dynatrace problem to the ServiceNow ticket. The correlation ID is what lets a later **Search incidents** task find the incident number that **Comment** and **Resolve** require (§5).

The integration is itself a production dependency, so monitor it like one — the queries below track execution volume, success rate, and failures so a broken connection surfaces as a metric, not as a silent gap in the incident queue.

Monitor Incident Integration

```
// Incident-management connector task executions
// Data object corrected 08/12/2026. Workflow executions are NOT in `events`,
and there is no
// `automation.task.execution` / `automation.workflow.execution` event type in
any spelling — those
// filters matched nothing, silently, in 25 cells. AutomationEngine writes to
`dt.system.events` with
// `event.kind == "WORKFLOW_EVENT"` (5.7M records / 30d here), split by
`event.type`:
//   WORKFLOW_EXECUTION · TASK_EXECUTION · ACTION_EXECUTION ·
WORKFLOW_CREATED/UPDATED/DELETED
// Fields are the `dt.automation_engine.*` family:
//   .workflow.title / .workflow.id      (was workflow.name)
//   .task.name                          (was task.name)
//   .action.function / .action.app      (was task.type — e.g. run-javascript,
send-email,
//                                        snow-search-incidents)
//   .state                              (was task.status / execution.status)
//                                        values RUNNING · SUCCESS · ERROR ·
DISCARDED · SKIPPED
//                                        — note "FAILED" is NOT a value; it
is ERROR
//   .state.is_final                    true only on terminal records —
filter on it, otherwise a
//                                        single execution is counted once as
RUNNING and again as
//                                        SUCCESS/ERROR
//   .state_info                         (was task.error)  ·  duration (was
task.duration)
// Enumerate with:
//   fetch dt.system.events, from:-24h | filter event.kind == "WORKFLOW_EVENT"
| limit 1
fetch dt.system.events, from:-7d
| filter event.kind == "WORKFLOW_EVENT"
| filter event.type == "ACTION_EXECUTION"
| filter dt.automation_engine.state.is_final == true
| filter contains(dt.automation_engine.action.app, "servicenow") or
contains(dt.automation_engine.action.app, "pagerduty") or
contains(dt.automation_engine.action.app, "jira")
| summarize executions = count(), by:{dt.automation_engine.action.app,
dt.automation_engine.action.function, dt.automation_engine.state}
| sort executions desc
```

```
// Failed incident management tasks
// Data object corrected 08/12/2026. Workflow executions are NOT in `events`, and there is no
// `automation.task.execution` / `automation.workflow.execution` event type in any spelling — those
// filters matched nothing, silently, in 25 cells. AutomationEngine writes to `dt.system.events` with
// `event.kind == "WORKFLOW_EVENT"` (5.7M records / 30d here), split by `event.type`:
//   WORKFLOW_EXECUTION · TASK_EXECUTION · ACTION_EXECUTION · WORKFLOW_CREATED/UPDATED/DELETED
// Fields are the `dt.automation_engine.*` family:
//   .workflow.title / .workflow.id      (was workflow.name)
//   .task.name                          (was task.name)
//   .action.function / .action.app      (was task.type — e.g. run-javascript, send-email,
//                                        snow-search-incidents)
//   .state                              (was task.status / execution.status)
//                                        values RUNNING · SUCCESS · ERROR · DISCARDED · SKIPPED
//                                        — note "FAILED" is NOT a value; it is ERROR
//   .state.is_final                     true only on terminal records — filter on it, otherwise a
//                                        single execution is counted once as RUNNING and again as
//                                        SUCCESS/ERROR
//   .state_info                         (was task.error)  ·  duration (was task.duration)
// Enumerate with:
//   fetch dt.system.events, from:-24h | filter event.kind == "WORKFLOW_EVENT" | limit 1
fetch dt.system.events, from:-24h
| filter event.kind == "WORKFLOW_EVENT"
| filter event.type == "ACTION_EXECUTION"
| filter dt.automation_engine.state == "ERROR"
| filter contains(dt.automation_engine.action.app, "servicenow") or contains(dt.automation_engine.action.app, "pagerduty")
| fields timestamp, dt.automation_engine.action.function, dt.automation_engine.state_info
| sort timestamp desc
| limit 25
```

```
// Connector activity per day
// Data object corrected 08/12/2026. Workflow executions are NOT in `events`, and there is no
// `automation.task.execution` / `automation.workflow.execution` event type in any spelling — those
// filters matched nothing, silently, in 25 cells. AutomationEngine writes to `dt.system.events` with
// `event.kind == "WORKFLOW_EVENT"` (5.7M records / 30d here), split by `event.type`:
//   WORKFLOW_EXECUTION · TASK_EXECUTION · ACTION_EXECUTION · WORKFLOW_CREATED/UPDATED/DELETED
// Fields are the `dt.automation_engine.*` family:
//   .workflow.title / .workflow.id      (was workflow.name)
//   .task.name                          (was task.name)
//   .action.function / .action.app      (was task.type — e.g. run-javascript, send-email,
//                                        snow-search-incidents)
//   .state                              (was task.status / execution.status)
//                                        values RUNNING · SUCCESS · ERROR · DISCARDED · SKIPPED
//                                        — note "FAILED" is NOT a value; it is ERROR
//   .state.is_final                     true only on terminal records — filter on it, otherwise a
//                                        single execution is counted once as RUNNING and again as
//                                        SUCCESS/ERROR
//   .state_info                         (was task.error)  ·  duration (was task.duration)
// Enumerate with:
//   fetch dt.system.events, from:-24h | filter event.kind == "WORKFLOW_EVENT" | limit 1
fetch dt.system.events, from:-30d
| filter event.kind == "WORKFLOW_EVENT"
| filter event.type == "ACTION_EXECUTION"
| filter dt.automation_engine.state.is_final == true
| filter contains(dt.automation_engine.action.app, "servicenow") or contains(dt.automation_engine.action.app, "pagerduty")
| makeTimeseries executions = count(), interval:24h
```

Next Steps

With incident management configured, customize message templates:

Recommended Path

1. **WFLOW-06: Custom Notification Templates** - Rich formatting
2. **WFLOW-07: Problem-Triggered Remediation** - Auto-remediation
3. **WFLOW-08: JavaScript & HTTP Actions** - Custom integrations

Key Takeaways

- **PagerDuty**: **Send event** triggers, acknowledges and resolves through Events API v2 (routing key); **Create an incident** uses the REST API (API key)
- **ServiceNow** uses a native connection (Basic or OAuth Client Credentials) with operations selected by name; `Category`, `Subcategory`, `Impact`, `Urgency`, `Assignment Group` are required on create
- **Deduplication**: PagerDuty dedups on `dedupKey`; for ServiceNow, **Search incidents** on the correlation ID before creating, because Create Incident does not dedupe
- **Lifecycle workflows** handle open/update/close events
- **Bi-directional**: the native action is one-way; true state sync comes from the ServiceNow-side Dynatrace app / ITOM (see ALERT-04)
- **Production hardening** — retry, dead-letter, link-back, and monitoring keep the integration trustworthy

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Summary

In this notebook, you learned:

- Integration benefits and architecture
- PagerDuty connection setup and tasks
- ServiceNow connection setup and tasks
- Bi-directional sync patterns
- Incident lifecycle management
- Deduplication strategies
- Production hardening (failure handling, dead-letter, monitoring)

References

- ServiceNow action (DT docs)
- Set up PagerDuty Connector (DT docs)
- Build workflows — task conditions and options (DT docs)
- Problem and event triggers (DT docs)
- PagerDuty action (DT docs)
- Send Dynatrace notifications to ServiceNow (DT docs)
- Notification actions umbrella (DT docs)
- HTTP request action (DT docs)
- Davis Problems app (DT docs)
- PagerDuty Events API v2 (PagerDuty Developer)
- ServiceNow REST API reference (ServiceNow Developer)

This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.